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United States Patent Application Publication 20250276855
Kind Code A1
Publication Date September 04, 2025
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GUARD APPARATUS FOR A CONVEYOR SYSTEM

Abstract

Various embodiments are directed to a guard apparatus for a conveyor system. The conveyor system comprises a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. The guard apparatus comprises a top cover and a first flange. The top cover is disposed in a first gap formed between a first roller and a second roller of the plurality of rollers. The first flange extends from a first end of the top cover into a second gap formed between a first end face of the first roller, a second end face of the second roller and a corresponding side frame of the conveyor system.

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Family ID: **94633576**

Appl. No.: **18/591544**

Filed: **February 29, 2024**

Publication Classification

Int. Cl.: **B65G39/12** (20060101)

U.S. Cl.:

CPC **B65G39/12** (20130101); **B65G13/11** (20130101); **B65G47/66** (20130101); B65G13/00 (20130101); B65G2207/40 (20130101)

Background/Summary

FIELD OF THE INVENTION

[0001] Various embodiments described herein relate generally to a material handling system for handling objects and, more particularly, to apparatuses for use with a conveyor system configured to facilitate transportation of objects along a conveyor surface.

BACKGROUND

[0002] Generally, in material handling environments like, but not limited to, distribution centers, warehouses, inventories, or shipping centers, a material handling system can convey, handle, sort, and organize various type of articles (for example objects, cartons, cases, containers, shipment boxes, totes, packages, and/or the like) using one or more conveyor systems. Through applied effort, ingenuity, and innovation, many of these identified problems have been solved by developing solutions that are included in embodiments of the present disclosure, many examples of which are described in detail herein.

SUMMARY

[0003] Various embodiments are directed to a guard apparatus for a conveyor system. The conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. The guard apparatus includes a top cover and a first flange. The top cover is disposed in a first gap formed between a first roller and a second roller of the plurality of rollers. The first flange extends from a first end of the top cover into a second gap formed between a first end face of the first roller, a second end face of the second roller and a corresponding side frame of the conveyor system.

[0004] In various embodiments, the first flange includes a central arm, a first arm and a second arm. The central arm extends from the first end of the top cover. The first arm extends from the central arm in a first direction. The second arm extends from the central arm in a second direction such that the second direction is opposite to the first direction.

[0005] In various embodiments, the first arm extends into the second gap between the first end face of the first roller and the corresponding side frame of the conveyor system. In various embodiments the second arm extends into the second gap between the second end face of the second roller and the corresponding side frame of the conveyor system.

[0006] In various embodiments, the central arm, the first arm and the second arm form a T-shape.

[0007] In various embodiments, the first flange is configured to secure the top cover between the first roller and the second roller.

[0008] In various embodiments, the conveyor system includes a driving member configured to connect the first roller with the second roller and transmit motion from the first roller to the second roller.

[0009] In various embodiments, a top surface of the top cover is disposed below the driving member.

[0010] In various embodiments, the driving member is a rigid belt.

[0011] In various embodiments, each of a first edge and a second edge opposite to the first edge of the top cover is tapered to confirm with the first roller and the second roller, respectively.

[0012] In various embodiments, the top cover defines a plurality of friction-reducing features proximal to at least one roller of the conveyor system.

[0013] In various embodiments, the guard apparatus further includes a second flange extending from a second end of the top cover. In various embodiments, the guard apparatus includes at least one support rib extending between the first flange and the second flange.

[0014] In various embodiments, the second flange is configured to extend into the first gap formed between the first roller and the second roller of the plurality of rollers.

[0015] In various embodiments, the second flange includes a first arcuate side and a second arcuate side, such that the first arcuate side and the second arcuate side conforms with the first roller and the second roller, respectively.

[0016] In various embodiments, the first flange defines a first length. In various embodiments, the second flange defines a second length. In various embodiments, the second length is less than the first length.
[0017] In various embodiments, the top cover, the first flange and the second flange are formed integrally as a single component to form a guard apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0018] The description of the illustrative embodiments can be read in conjunction with the accompanying figures. It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements are exaggerated relative to other elements. Same numbers are used throughout the figures to reference like features and components. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the figures presented herein, in which:
[0019] FIG. **1**A illustrates a top view of a conveyor system, in accordance with various embodiments of the present disclosure;
[0020] FIG. **1**B illustrates a front perspective view of a guard apparatus and a plurality of rollers of the conveyor system, in accordance with various embodiments of the present disclosure;
[0021] FIG. **2**A illustrates a top perspective view of the guard apparatus of FIG. **1**B, in accordance with various embodiments of the present disclosure; and
[0022] FIG. **2**B illustrates a bottom perspective view of the guard apparatus of FIG. **2**A, in accordance with various embodiments of the present disclosure.
DETAILED DESCRIPTION
[0023] Indeed, various embodiments may be implemented in many different forms and should not be construed as limited to the embodiments set forth herein. Other embodiments, and modifications and improvements of the described embodiments, will occur to those skilled in the art and all such other embodiments, modifications and improvements are within the scope of the present disclosure. Features from one embodiment or aspect may be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments may be applied to apparatus, product or component aspects or embodiments and vice versa. Like numbers refer to like elements throughout.
[0024] It should be understood at the outset that, although illustrative implementations of one or more aspects are illustrated below, the disclosed assemblies and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and techniques illustrated and/or described below, but may be modified within the scope of the appended claims along with their full scope of equivalents. While values for dimensions of various elements are disclosed, the drawings may not be to scale.
[0025] The words “example,” or “exemplary,” when used herein, are intended to mean “serving as an example, instance, or illustration.” Any implementation described herein as an “example” or “exemplary embodiment” is not necessarily preferred or advantageous over other implementations. The phrases “in an embodiment,” “in some embodiments,” “according to one embodiment,” “in various embodiments” and the like generally mean that a particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure or may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).
[0026] As used herein, directional terms used to describe a component, configuration, action,

position, direction, and/or the like (for example, “rearward,” “an upward vertical direction,” “laterally outward,” “bottom,” “top,” and the like) are meant to be interpreted relative to a hypothetical conveyor system provided on an at least substantially flat horizontal surface, but the terms are not to be interpreted as requiring the conveyor system to be in this orientation at any particular time.

[0027] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0028] Conveyor systems are used in, for example, industrial manufacturing and packaging applications to facilitate the transportation of objects to a desired downstream location within a factory or a warehouse. For example, the conveyor systems include a plurality of cylindrical rollers disposed parallel and adjacent to each other along a length of the conveyor system. The plurality of rollers is arranged relative to one another such that the respective rolling surfaces of the rollers collectively define a conveyor path along which an object may be, at least, continuously transported towards the downstream location. Each roller rotates about an axis thereof to help transfer objects disposed thereon to an adjacent roller and, in doing so, together, the plurality of cylindrical rollers aids in transferring the objects along a conveyor travel path towards an end of the conveyer system. Further, the conveyor systems include conveyor frame(s) having various structural components, for example, but not limited to, sidewalls, panels, and/or the like, which are assembled relative to one another to define extremes capable of supporting the plurality of rollers therebetween.

[0029] In conveyor systems, the motion of the rollers poses a safety concern for an operator, such as, one who handles the items being transferred on the conveyor system. There is a risk of accidental finger injury of the operator handling the objects of the conveyor system. In some examples, a guard apparatus may be installed within the conveyor system, by being secured relative to the side frames of the conveyor system, to protect against physical injury of the operator. The severity of injury to the operator's finger stuck in a gap between the moving rollers and the stationary side frame may be high. Conventional guard apparatuses are not effective to prevent such incidents.

[0030] The guard apparatus includes one or more stationary cover portions configured to cover the gap between two adjacent rollers of the conveyor system. Although such guard apparatuses are used to prevent unauthorized and/or unintended access to an internal portion, they fail to laterally secure the guard apparatus between the side frames. Additionally, the guard apparatus is subjected to repeated loading cycles, leading to localized deformations. Such deformations, in the absence of lateral security, may cause a loosening of the guard apparatus installed within the conveyor system, which may result in undesirable movement thereof relative to the conveyor frame, undesirable noise (for example, rattling), partial inoperability of the guard apparatus, and/or unintentional disassembly of the guard apparatus relative to the conveyor system.

[0031] Various embodiments of the present disclosure are directed to a guard apparatus for a conveyor system. FIG. **1**A illustrates a top view of a conveyor system **100**, in accordance with various embodiments of the present disclosure. The conveyor system **100** includes a plurality of rollers **102** (such as a first roller **102-1**, a second roller **102-2**, a third roller **102-3**, and a fourth roller **102-4**), hereinafter alternatively, individually, and collectively referred to as “the roller(s) **102**” or “adjacent rollers **102-1**, **102-2**”, disposed between a pair of side frames **104-1**, **104-2** (hereinafter alternatively and collectively referred to as “the side frames **104**”), such that a first gap **106** exists between any two adjacent rollers **102-1** and **102-2**.

[0032] As shown in FIG. **1**A, each roller **102** defines an end face (for example, the first roller **102-1** defines a first end face **108-1**, the second roller **102-2** defines a second end **108-2**) hereinafter alternatively, individually, and collectively referred to as “the end face(s)”. In various embodiments, a second gap **110** is formed between the first end face **108-1** of the first roller **102-1**, the second end face **108-2** of the second roller **102-2** and a corresponding side frame **104-1**.

[0033] The adjacent rollers **102-1**, **102-2** are joined by one or more driving members (**112**), such as a belt or an O-ring at one end, where the driving member **112** is configured to transmit a rotary motion from one roller to the other, without slippage. The conveyor system **100** may include multiple such driving members configured to connect each pair of adjacent rollers **102-1**, **102-2** to transmit the motion throughout a length of the conveyor system **100** in order to facilitate transfer of objects.

[0034] As seen in FIG. **1**A and FIG. **1**B, a guard apparatus **114** is positioned within the conveyor system **100** to provide coverage over at least a portion of the first gap **106** between two adjacent rollers **102-1**, **102-2**. In some embodiments, the guard apparatus **114** further extends into at least a portion of the second gap **110** between the first end face **108-1** of the first roller **102-1**, the second end face **108-2** of the second roller **102-2**, and the corresponding side frame **104-1** of the conveyor system **100**.

[0035] Referring to FIG. **2**A a top perspective view of the guard apparatus **114** is illustrated. The guard apparatus **114** includes a top cover **202** and a first flange **204**. The top cover **202** is disposed in the first gap **106** (See FIG. **1**A) between the first roller **102-1** and the second roller **102-2** of the plurality of rollers **102**. The top cover **202** includes a first end **208** and a second end **206**. The first flange **204** extends from the first end **208** of the top cover **202** into the second gap **110** formed between the first end face **108-2** of the first roller **102-1**, the second end face **108-2** of the second roller **102-2** and the corresponding side frame **104-1** of the conveyor system **100** (See FIG. **1**A and FIG. **1**B).

[0036] In various embodiments, the first flange **204** includes a central arm **210**, a first arm **212** and a second arm **214** (See FIG. **2**B). The central arm **210** extends from the first end **208** of the top cover **202** into the second gap **110**. The first arm **212** extends from the central arm **210** in a first direction and the second arm **214** extends from the central arm **210** in a second direction, such that the second direction is opposite to the first direction. In various embodiments, the first arm **212** particularly extends between the second end face **108-2** of the second roller **102-2** and the corresponding side frame **104-1** of the conveyor system **100**. In various embodiments, the second arm **214** particularly extends between the first end face **108-1** of the first roller **102-1** and the corresponding side frame **104-1** of the conveyor system **100**. In various embodiments, the central arm **210**, the first arm **212** and the second arm **214** forms a T-shape. In various embodiments, the first arm **212** and the second arm **214** are configured to contact at least a portion of the first end face **108-1** and the second end face **108-2** respectively. In various embodiments, edges of the first arm **212** and the second arm **214** are configured to contact a roller support element (not shown). In various embodiments, the first flange **204** is configured to secure the top cover **202** between the first roller **102-1** and the second roller **102-2**. In various embodiments, a top surface **220** of the top cover **202** is disposed below the driving member **112**.

[0037] In various embodiments, each of a first edge **222** and a second edge **224** opposite to the first edge **222** of the top cover **202** is tapered to confirm with the first roller **102-1** and the second roller **102-2**, respectively. In various embodiments, the top cover **202** defines a plurality of friction-reducing features **226** (See FIG. **2**B) proximal to at least one roller **102-1**, **102-2** of the conveyor system **100**. In various embodiments, the friction-reducing features **226** include a plurality of tapered grooves. In various embodiments, the friction-reducing features **226** are equidistantly spaced-apart from each other. In various embodiments, the top surface **220** is enclosed by the first end **208**, the second end **206**, the first edge **222** and the second edge **224** of the top cover **202**.

[0038] In various embodiments, the guard apparatus **114** further includes a second flange **228** extending from the second end **206** of the top cover **202**. In various embodiments, the guard apparatus **114** includes at least one support rib **230** (See FIG. **2**B) extending between the first flange **204** and the second flange **228**. In various embodiments, the second flange **228** is configured to extend into the first gap **106** formed between the first roller **102-1** and the second roller **102-2** of the plurality of rollers **102**. In various embodiments, the second flange **228** includes a first arcuate side **232** and a second arcuate side **234**, such that the first arcuate side **232** and the second arcuate side **234** conforms with the first roller **102-1** and the second roller **102-2**, respectively.

[0039] In various embodiments, the first flange **204** defines a first length “L1”. In various embodiments, the second flange **228** defines a second length “L2”. In various embodiments, the second length “L2” is less than the first length “L1”. In various embodiments, the guard apparatus **114** may define a stiffening groove **240** on the top cover **202** to strengthen the guard apparatus **114**.

[0040] In various embodiments, the guard apparatus **114** may be made up of light-weight polymer, such as plastics manufactured through, for example, injection molding. To this end, the top cover **202**, the first flange **204** and the second flange **228** are formed integrally as a single component to form a guard apparatus **114**.

[0041] Shape of the illustrated guard apparatus **114** should not be construed as limited. Instead, other shapes and features will be apparent to the person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, the guard apparatus **114** may include an inverted U-shaped elongated wall with extending side flanges as a part of the top cover **202**. Further, the guard apparatus **114** may include multiple fillets, chamfer, or edge radii.

[0042] The present disclosure provides an improved guard apparatus **114** for the conveyor system **100**. The guard apparatus **114** provides an effective protection against injury as it is configured to cover the gap formed between stationary members (such as side frames **104-1**, **104-2**) and movable members (such as rollers **102-1**, **102-2**) of the conveyor system **100**. The guard apparatus **114** as per the present disclosure also prevents unauthorized and/or unintended access to portion of conveyor system **100**. Moreover, the guard apparatus **114** also prevents accumulation of debris, or small article, such as fasteners or coins, which may lead to jamming of the conveyor system **100**.

[0043] Various embodiments of the present disclosure provide further advantages. As mentioned hereinafter, the term “lateral” or “laterally” is used to refer to the direction along length of the roller of the conveyor system and the term “longitudinal” or “longitudinally” is used to refer to the direction along which an object moves along while the conveyor system is in an operating condition. Owing to the first flange **204**, the guard apparatus **114** is secured laterally as the first arm **212** and the second arm **214** are configured to be engaged with the first end face **108-1** and the second end face **108-2** of the first roller **102-1** and the second roller **102-2** respectively. Additionally, owing to the first arcuate side **232** and the second arcuate side **234** conforming to the adjacent rollers **102-1**, **102-2**, the guard apparatus **114** is longitudinally secured. Moreover, the first arm **212** and the second arm **214** of the first flange **204** are configured to contact a roller support clement and provide a vertical securing of the guard apparatus **114**. The lateral, longitudinal and vertical security of the guard apparatus **114** reduces undesirable noise (for example, rattling) and damage of the guard apparatus **114**, thereby increasing the usable life of the guard apparatus **114**.

[0044] It may be desirable to manufacture the guard apparatus **114** using simpler methods, such as injection molding. The present disclosure provides a simplified structure for the guard apparatus **114** thereby reducing the manufacturing costs, etc. Owing to the design of the guard apparatus **114**, the elements of the guard apparatus **114** are formed integrally as a single component as the top cover **202**, the first flange **204** and the second flange **228** are formed integrally as a single component to form the guard apparatus **114**. This reduces the number of components and the cost of installation of the guard apparatus **114** on the conveyor system **100**.

[0045] Owing to the stiffening groove **240** on the top cover **202**, the strength of the top cover **202** is improved without usage of additional or separate stiffening elements, such as metal fasteners, or metal support plate, etc. Additionally, owing to the support rib **230** extending between the first flange **204** and the second flange **228**, the bending strength of the top cover **202** is enhanced. An improvement in strength and stiffness of the top cover **202** leads to reduced localized deformations of the guard apparatus **114** due to, for example, repeated or cyclic loading.

[0046] Owing to the friction-reducing features **226** of the top cover **202**, the frictional forces between the stationary top cover **202** of the guard assembly **114** and the adjacent rollers **102-1**, **102-2** of the conveyor system **100** (such as a first roller **102-1** and a second roller **102-2**) may be significantly reduced thereby increasing the usable life of the guard apparatus **114**.

[0047] When an element or layer is referred to as being "on," "engaged to," "connected to," or "coupled to" another element or layer, it may be directly on, engaged, connected, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (for example, "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0048] Spatially relative terms, such as "inner," "outer," "beneath," "below," "lower," "above," "upper," and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures.

[0049] Many modifications and other embodiments of the present disclosure will be apparent to one skilled in the art to which the present disclosure pertain having the benefit of teachings presented in the foregoing descriptions and the associated drawings. Although the figures only show certain components of the system described herein, it is understood that various other components may be present. Therefore, it is to be understood that the present disclosure should not be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, the steps in the method described above may not necessarily occur in the order depicted in the accompanying diagrams, and in some cases one or more of the steps depicted may occur substantially simultaneously, or additional steps may be involved. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A guard apparatus for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the guard apparatus comprising: a top cover disposed in a first gap formed between a first roller and a second roller of the plurality of rollers; and a first flange extending from a first end of the top cover into a second gap formed between a first end face of the first roller, a second end face of the second roller and a corresponding side frame of the conveyor system.

2. The guard apparatus of claim 1, wherein the first flange comprises: a central arm extending from the first end of the top cover; a first arm extending from the central arm in a first direction; and a second arm extending from the central arm in a second direction, wherein the first direction is opposite to the second direction.

3. The guard apparatus of claim 2, wherein the first arm extends into the second gap between the first end face of the first roller and the corresponding side frame of the conveyor system, and wherein the second arm extend into the second gap between the second end face of the second roller and the corresponding side frame of the conveyor system.

4. The guard apparatus of claim 2, wherein the central arm, the first arm and the second arm form a T-shape.

5. The guard apparatus of claim 1, wherein the first flange is configured to secure the top cover

between the first roller and the second roller.
6. The guard apparatus of claim 1, further comprising a driving member configured to: connect the first roller with the second roller; and transmit motion from the first roller to the second roller.
7. The guard apparatus of claim 6, wherein a top surface of the top cover is disposed below the driving member.
8. The guard apparatus of claim 6, wherein the driving member is a rigid belt.
9. The guard assembly of claim 1, wherein each of a first edge and a second edge opposite to the first edge of the top cover is tapered to conform with the first roller and the second roller, respectively.
10. The guard apparatus of claim 1, wherein the top cover defines a plurality of friction-reducing features proximal to at least one roller of the conveyor system.
11. The guard apparatus of claim 1, further comprising: a second flange extending from a second end of the top cover; and at least one support rib extending between the first flange and the second flange.
12. The guard apparatus of claim 11, wherein the second flange is configured to extend into the first gap formed between the first roller and the second roller of the plurality of rollers.
13. The guard apparatus of claim 11, wherein the second flange comprises a first arcuate side and a second arcuate side, each of the first arcuate side and the second arcuate side conforming with first roller and the second roller, respectively.
14. The guard apparatus of claim 11, wherein a first length of the first flange is more than a second length of the second flange.
15. The guard apparatus of claim 11. wherein the top cover, the first flange and the second flange are formed integrally as a single component.